
Layer-wise Analysis of Internal VLM Features via Logit Lens on Qwen2-VL-2B-Instruct

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1. Introduction

I ran the logit lens on **Qwen2-VL-2B-Instruct** across five feature categories—color, shape, counting, spatial relations, and attribute-object binding—on 1250 controlled synthetic images. Color, shape, and compositional binding all share a sharp two-layer transition at L22→L23 rather than a gradual rise; binding shows up one layer after unbound color, with the distractor color already pinned near zero. Counting falls apart for $n \geq 4$ —the model floors at “three”—and even count “one” has to fight a plural prior until L27. “Below” is the strangest case: P(below) rises through L24, then *decays*, and every L28 error produces “right.” [Neo et al. \(2025\)](#) previously reported best object-identity decoding around layer 25 of 28 on this same backbone; this paper extends the per-layer breakdown to the four other feature classes.

2. Method

2.1. Model

Qwen2-VL-2B-Instruct: 28 decoder layers, hidden size 1536, pre-norm RMSNorm, fp16 on a T4. The lens is well-defined:

$$p_\ell = \text{softmax}(W_U \cdot \text{RMSNorm}(h_\ell)) \quad (1)$$

h_ℓ is the residual-stream hidden state at layer ℓ and W_U is the unembedding matrix.

2.2. Features and Dataset

Five synthetic datasets, all 448×448 px on a light gray (#D0D0D0) background, reproducible per-image seeds.

Color (200 images). A single circle ($r=60$ px) in one of four colors (red, blue, green, yellow), centered with ± 30 px jitter.

Shape (200 images). Four shapes (circle, square, triangle, 5-pointed star), white fill with black outline, jittered around center. Color held constant.

Counting (250 images). $N \in \{1, \dots, 5\}$ red circles ($r=25$ px) at random non-overlapping positions.

Spatial relations (200 images). A red circle and a blue square in four configurations (left, right, above, below). Positions are axis-aligned so the only signal differentiating “above” from “below” is the vertical coordinate—an earlier non-aligned version gave the model a horizontal cue and confused it.

Attribute-object binding (400 images). Contrastive pairs. Image A: red circle on the left, blue square on the right; image B swaps the colors. The question “What color is the circle?” should produce “red” for A and “blue” for B. 200 pairs test whether the model binds color to the queried object or just registers “red” and “blue” both being present.

2.3. Prompt Design

Every prompt prefills the assistant turn so the answer lands at a known token position. Color, for example:

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USER: <image> What color is the  
object? Answer with one word.  
ASSISTANT: The color is
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The model’s next token after “is” is the answer. Every target word has a single-token surface form in the Qwen2 tokenizer ($V=151,936$).

For counting, digit prompts give “ 4”—a space token followed by the digit—useless for a single-position lens, so I switched to word-form targets (“one” through “five”), each one token. For spatial relations, the only template (out of six I tried) that reliably gave a single-word completion was “Is the red circle above, below, left of, or right of the blue square? Answer with one word.” with prefix “Answer:”.

2.4. Metrics

At the answer position—last token of the prefix—and at each of the 29 hidden states (embedding + 28 layers) I record $P(\text{target})$ summed over single-token surface-form IDs, margin $(P(\text{target}) - \max_j P(\text{distractor}_j))$ across the

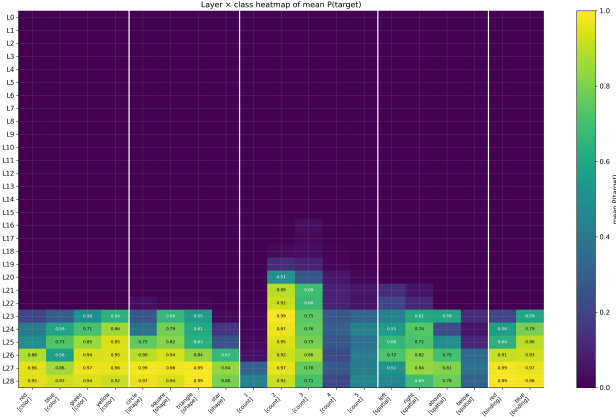


Figure 1. Layer $\times$ class heatmap of mean $P(\text{target})$. Cells with value > 0.5 are annotated. The L23 boundary cleanly separates zero signal from strong feature decoding for every working class. The “below” column and the count “one”/“four” columns stay dark across all 29 rows.

Table 1. Feature emergence: mean $P(\text{target})$ at key layers and L28 accuracy ($\text{rank}_{\text{target}}=0$). Emergence layer = first layer with mean $P(\text{target}) > 0.1$.

Category	L21	L23	L25	L28	Acc.
Color	0.000	0.413	0.755	0.936	100%
Shape	0.001	0.374	0.617	0.925	100%
Binding	0.000	0.400 [†]	0.770 [†]	0.987	100%
Count	0.349	0.522	0.484	0.548	71%
Spatial	0.046	0.307	0.473	0.562	78%

[†]Binding values are margin ($P(\text{target}) - P(\text{distractor})$), not raw $P(\text{target})$.

other classes), rank, and full-vocabulary entropy. Aggregates use bootstrap 95% CIs (1000 resamples).

3. Results

The full sweep is $1250 \text{ images} \times 29 \text{ layers} = 36,250$ measurements.

3.1. Phase Transition at L22–L23

Figure 1 plots mean $P(\text{target})$ across every layer and class. There’s a hard boundary at L22–L23: below it, nothing is decodable; above it, every working feature is already strong. The transition takes 2–3 residual blocks, not a smooth ramp. Table 1 has the numbers. For color, $P(\text{target})$ goes from 0.001 at L22 to 0.413 (95% CI [0.391, 0.437]) at L23—a $400\times$ jump in one layer. Shape and binding hit the same window.

3.2. Color and Shape

Color and shape are the cleanest features. All four colors emerge together at L23 ($P > 0.2$ each, no privileged hue—Figure 2, leftmost panel), and by L26 every one is rank-0 with $P > 0.88$. At L28, per-class accuracy is 100% across the board; red sits highest at $P = 0.955$, yellow lowest at $P = 0.921$, and the gap is well within the bootstrap CI.

Shape works on the same schedule with one small lag—the 5-pointed star arrives a layer late (L24 vs. L23 for circle, square, triangle) and plateaus at $P = 0.798$ instead of the 0.94–0.99 the other three reach. My guess is the white-on-gray star activates a thinner visual prior than the solid filled shapes, but the lens alone can’t prove that.

3.3. Binding: Compositional, Not Bag-of-Words

The binding test is whether the model attaches “red” to “circle” (compositional) or just registers that both “red” and “circle” apply somewhere in the image (bag-of-words). On this dataset the answer is compositional—Figure 3.

Panel (b) is the one to look at: $P(\text{distractor})$ stays below 0.01 at *every* layer. The model never goes through a phase of “I see red but I don’t know whether it belongs to the circle or the square.” When the signal first appears, it’s already bound. Panel (d) is the harder check—pair-level accuracy (both A and B correct on the same pair) is exactly 0% through L23 and 100% from L24, which kills the “the model just always says one color” explanation. You can’t get 100%/100% single-side accuracy on both A and B by predicting the same color twice.

Binding emerges one layer after unbound color (L24 vs. L23). Whether that one-layer gap is a real compositional circuit or just later softmax sharpening, the lens can’t tell—you’d need patching or activation ablation.

3.4. Counting: Subitizing Ceiling at Three

Counting is the first feature that doesn’t work. Figure 4 splits cleanly: “two” and “three” saturate by L21–L23 ($P > 0.85$), while “four” and “five” plateau at 0.2–0.45 and never recover.

Two distinct failure modes:

$n = 4$ **hard-floors at “three.”** Of the 39 wrong predictions on $n=4$ images, every single one is “three.” Not noise—a deterministic preference. On $n=5$, 82% of errors are “four” and the rest “three,” which looks like a one-step undercount baked in.

$n = 1$ **loses to a language prior.** Count “one” emerges only at L27, vs. L19 for “two”—which is jarring if you expect “one” to be the easy case. The “There are” prefill bi-

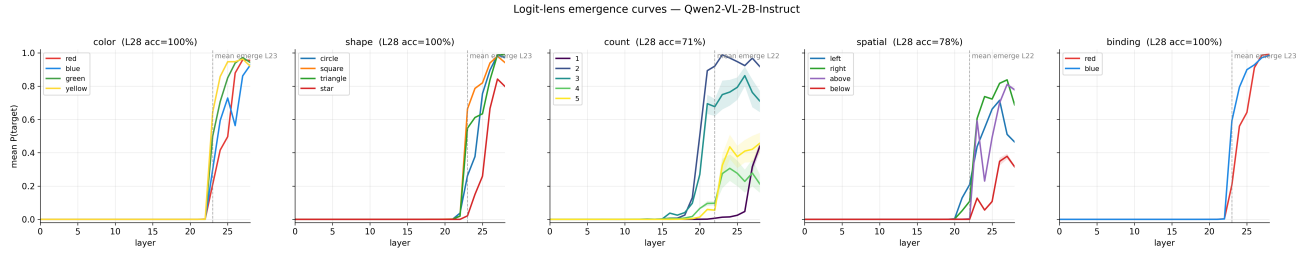


Figure 2. Logit-lens curves: mean $P(\text{target})$ vs. layer for the five categories, broken out by class. Dashed vertical lines mark the per-category emergence threshold (mean $P > 0.1$). Color and shape share the same sharp L23 transition; counting splits, with “two” at L19 and “one” not appearing until L27; “below” never settles; binding differentiates A vs. B from L23.

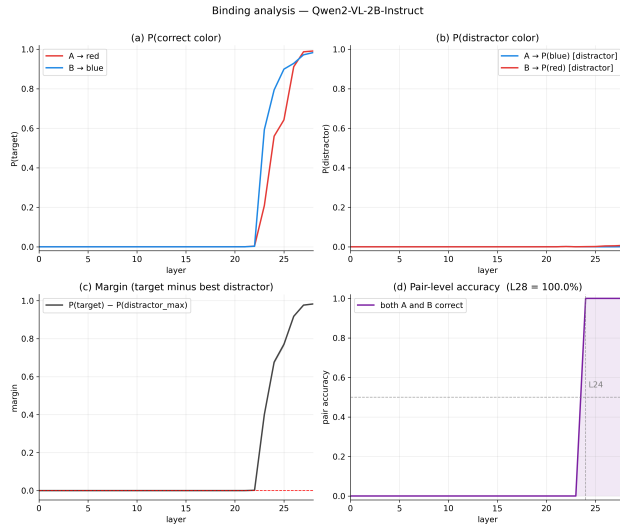


Figure 3. Binding analysis. (a) $P(\text{correct color})$ for $A \rightarrow \text{red}$ and $B \rightarrow \text{blue}$. (b) $P(\text{distractor color})$ —stays at zero throughout, killing the bag-of-words alternative. (c) Margin (target minus best distractor). (d) Pair-level accuracy: 0% through L23, 100% from L24.

ases toward plural continuation, so for L19 through L26 the model is voting “two” or “three” even on single-object images. The visual evidence eventually wins, but only at the very last layer. That’s eight layers of language-vision tug-of-war.

L28 per-class accuracy: 1→86%, 2→100%, 3→82%, 4→22%, 5→66%.

3.5. Spatial: The “Below” Blind Spot

Three of four spatial relations work—left, right, and above all hit 100% at L28. “Below” is broken: 10% accuracy with all 45 errors producing “right” (Figure 5).

The trajectory is the strange part. $P(\text{below})$ actually *rises* at L23–L24, then drops back through L28—the model has partial information about “below” at mid-depth and the late layers overwrite it with “right.” That is not absence of a

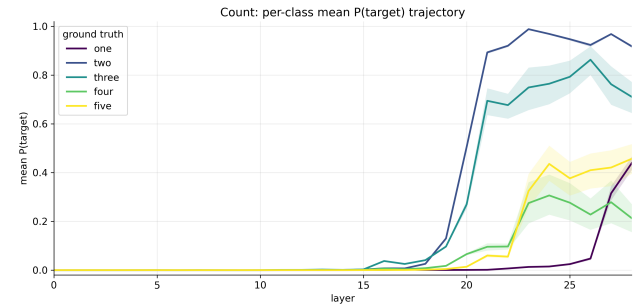


Figure 4. Count per-class trajectories. “Two” and “three” saturate by L21. “Four” and “five” plateau at 0.2–0.45. “One” only appears at L27.

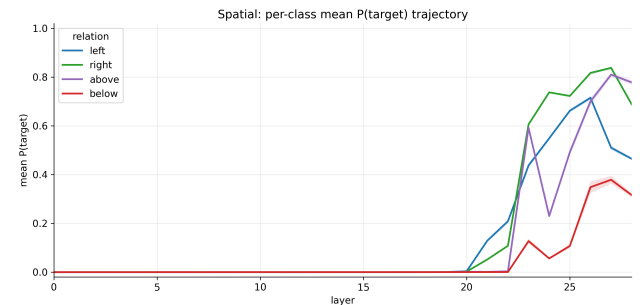


Figure 5. Spatial per-class trajectories. Left, right, and above climb out of zero at L21–L23. “Below” rises briefly at L23–L24, then *actively decays*—late layers suppress the correct answer.

feature; it is active suppression by something. I tried six prompt templates and an axis-aligned dataset redesign (zero perpendicular offset between objects), and got the same below→right collapse every time, matching the “weak spatial reasoning” line on Qwen2-VL’s own model card.

The working spatial classes are also globally weaker than color/shape—L28 $P(\text{target})$ is 0.47–0.78 here vs. 0.92–0.95 for the cleaner features—so spatial reasoning looks coarsely encoded across the board, not just on “below.”

4. Key Statements

Statement 1: Color and shape emerge in a sharp 2-layer phase transition at L22→L23. P(target color) goes from 0.001 at L22 to 0.413 [0.391, 0.437] at L23; shape from 0.019 to 0.374 [0.337, 0.408]. By L25 the target is rank-0 in every image. The whole transition lives in the last quarter of a 28-layer model—matching the L25/28 best-decoding layer that Neo et al. (2025) report on the same backbone.

Statement 2: Binding is compositional on this dataset—pair accuracy crosses 50% at L24 and saturates at 100%. Across 200 image pairs, P(distractor) never tops 0.01 and the L24 margin is +0.676 [0.665, 0.689]. The model knows which object the question is about as soon as it has anything to say—there is no “both colors are likely” midstream.

Statement 3 (non-working): Counting collapses at $n \geq 4$ —the model floors at “three.” P(four) at L28 on $n=4$ images is 0.213 [0.159, 0.271], while P(two) on $n=2$ reaches 0.918 [0.912, 0.924]. “Four” and “five” plateau after L23 and don’t recover at any later depth. That looks like a capacity ceiling rather than an optimization problem—the answer never appears in the residual stream.

Statement 4 (non-working): “Below” is actively suppressed in late layers. P(below) at L28 is 0.316; accuracy 10%; all 45 errors produce “right.” P(below) rises through L24 and then decays—late layers overwrite a partially formed spatial answer with a directional prior. Six prompt templates produce the same collapse, so this is a model-level blind spot, not a prompt artifact.

5. Limitations

Synthetic data. Solid shapes on flat gray backgrounds are very far OOD from this model’s pretraining mix. Real images would probably give smoother trajectories, possibly at different layers.

Correlational, not causal. The lens shows that a feature is *decodable* at layer ℓ , not that the model *uses* it there. To claim use you need patching or ablation.

Binding dataset is too clean. 100% pair accuracy and zero distractor probability come from a maximally easy setup—two well-separated objects, maximally distinct colors—not from a general VLM capability. Natural-image binding fails routinely (Yuksekgonul et al., 2023).

6. Conclusion

Color, shape, and compositional binding all emerge sharply at L22–L23; counting breaks at $n \geq 4$ and “below” gets overwritten in late layers. The whole story plays out in the last quarter of the network—layers 0–21 produce no decod-

able visual signal.

References

- Neo, C., Ong, L., Torr, P., Geva, M., and Khakzar, A. Towards interpreting visual information processing in vision-language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.07149.
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